

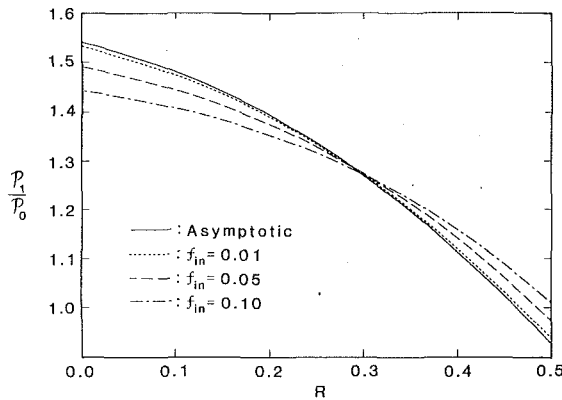


Fig. 3 Ratio of Φ_1 , given by equation (21), to the zeroth-order approximation of the extrusion pressure, Φ_0 , as a function of the area reduction R . The first-order approximation of the extrusion pressure is given by $\Phi \approx \Phi_0 + f_{in}\Phi_1$.

sion for Φ_1 , we first eliminate Y_1 from equation (17) using equations (15) and (18a) or (18b); the relation which results is

$$\Sigma_{r1}' = ab\sigma_y F \cosh Z_0. \quad (20)$$

Upon inserting the expression for F obtained from equation (16) into this relation and integrating, we find

$$\Phi_1 = \frac{2}{3}\sigma_y e^{-\beta} \int_{\alpha}^{\beta} \frac{e^w w}{\sqrt{w^2 - 1}} dw \quad (21)$$

where $\alpha = \cosh(1/b)$ and

$$\beta = \cosh[1/b - (3b/2)\ln(1 - R)]. \quad (22)$$

In Fig. 3, the extrusion pressures predicted by the first-order approximation are compared with the exact solution for cases corresponding to those in Fig. 2. The agreement is excellent for $f_{in} = 0.01$, and remains good even for the larger inlet void volume fractions. Note that in this figure it is Φ_1/Φ_0 which is plotted, so that even for the largest void volume fraction presented, the actual error in the approximate extrusion pressure is very small (about one percent).

Finally, we remark that the present analysis can be extended to include strain hardening of the matrix material. In particular, equation (16) generalizes to

$$\frac{F'}{F} = -\sqrt{3}^a \sinh\left(\frac{Z_0}{X_0^2 + Y_0^2}\right) \quad (23)$$

where $\{X_0, Y_0, Z_0\}$ are known from the corresponding radial flow solutions for void-free strain-hardening solids. For particular matrix-hardening characterizations (e.g., a pure power-law), the integration of equation (23) can be performed analytically.

Acknowledgments

A portion of this study has been supported by the L. Kraus Fund, Technion, Haifa, and by a grant from the ALCOA Foundation.

References

- Avitzur, B., and Blum, P., 1984, "Forging and Extrusion of P/M Preforms," *Mod. Dev. in Powder Metallurgy*, Vol. 8, H. H. Hausner and W. E. Smith, eds., Princeton, p. 73.
- Durban, D., 1979, "Axially Symmetric Radial Flow of Rigid/Linear-Hardening Materials," *ASME JOURNAL OF APPLIED MECHANICS*, Vol. 46, p. 322.
- Durban, D., 1983, "Radial Flow Simulation of Rigid/Hardening Materials," *Int. J. Mech. Sci.*, Vol. 25, p. 27.
- Durban, D., and Budiansky, B., 1979, "Plane Strain Radial Flow of Plastic Materials," *J. Mech. Phys. Solids*, Vol. 26, p. 303.
- Ferguson, B. L., 1984, "Forging and Rolling of P/M Billets," *Metals Handbook*, 9th ed., Vol. 7, p. 522.
- Gurson, A. L., 1977a, "Continuum Theory of Ductile Rupture by Void Nucleation and Growth—Part I. Yield Criteria and Flow Rules for Porous Ductile Media," *J. Engrg. Mater. Technol.*, Vol. 99, p. 2.

- Gurson, A. L., 1977b, "Porous Rigid-Plastic Materials Containing Rigid Inclusions—Yield Function, Plastic Potential, and Void Nucleation," *Fracture 1977*, Vol. 2, ICF4, Waterloo, Canada.
- Lenel, F. V., 1980, *Powder Metallurgy: Principles and Applications*, Metal Powder Industries Federation, Princeton, N.J., pp. 324–329; 456–464.
- Mear, M. E., and Durban, D., 1989, "Radial Flow of Sintered Powder Metals," *Int. J. Mech. Sci.*, Vol. 31, p. 37–49.

The Branched Crack Problem Revisited

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Introduction

In ceramic-matrix composites, preferential cracking of the matrix first occurs during monotonic loading. When the loading consists of tension applied parallel to the fibers, the resulting matrix cracks impinge upon the fiber-matrix interface. If the interface is sufficiently weak, as is often the case, debonding of the interface will occur prior to failure of the fibers. One is interested in the response of this branched crack under further increases in the load.

Since the branch-crack faces are just two sides of the fiber-matrix interface, it is clear that overlapping of these faces—material interpenetration—cannot be permitted. Yet, such interpenetration is a common feature of most solutions to branch-crack problems in homogeneous materials (see, for example, Vitek, 1977; Kitagawa and Yuuki, 1977; Lo, 1978), as evidenced by the negative mode I stress intensity factor at the tip of the branch crack. In this Note, we focus on the nature of branch-crack solutions when interpenetration is properly prohibited. This is done for the admittedly idealized problem of a doubly-symmetric branch crack in a homogeneous plane. Application of the notions developed here to the composite problem will be taken up in the near future.

Problem Statement and Analysis

A doubly-symmetrical branched crack subjected to remote tension σ_{∞} is shown schematically in Fig. 1. The main crack faces are assumed to be free of tractions. Since each branch is essentially identical for this loading, we will henceforth refer only to the branch along $x=0$, $0 < y < L$; analogous statements apply to the other branches. We assume that the branch-crack faces are fully lubricated so that the interface conditions along the branch crack are:

$$\sigma_{xy} = 0 \quad (1)$$

and

$$\sigma_{xx} = 0 \quad \text{if } h > 0 \quad (2a)$$

$$h = 0 \quad \text{if } \sigma_{xx} < 0 \quad (2b)$$

where h is the opening along the branch crack. It turns out that the interface will be open and will satisfy equation (2a) over a portion of the interface, $0 < y < d$, while the interface will be closed and will satisfy (2b) over $d < y < L$.

We employed a method of solution which is essentially identical to that of Lo (1978). The solution is expressed as a superposition of a single crack in an infinite plane plus a suitable

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Manuscript received by the ASME Applied Mechanics Division, Oct. 11, 1989, final revision, Apr. 26, 1990.

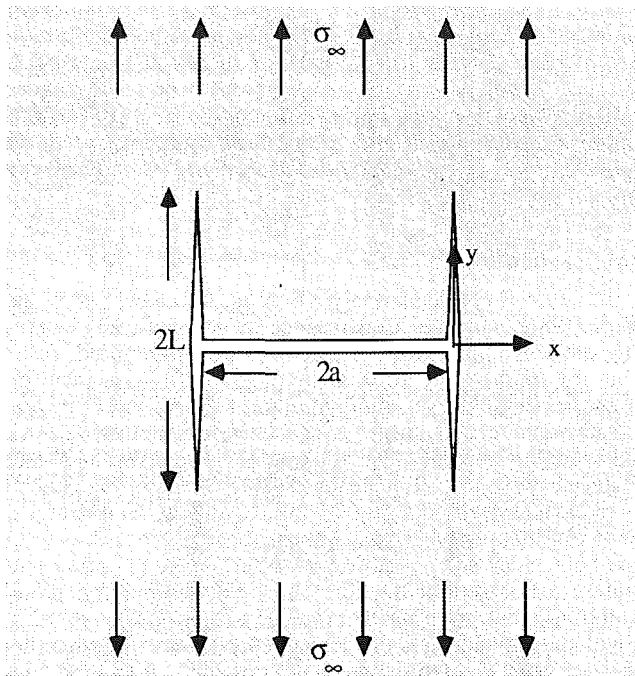


Fig. 1 Schematic of doubly-symmetrical branched crack

distribution of dislocations to represent the branch cracks. One needs to ensure that the main crack faces remain free of tractions. This is accomplished automatically by having *each* dislocation in the distribution leave the main crack faces traction-free. Thus, the appropriate kernel solution is the solution to a single dislocation in an infinite plane containing a traction-free crack lying along $-2a < x < 0$. This procedure yields a pair of singular integral equations for the dislocation densities $b_x(y)$ and $b_y(y)$, which we write in abbreviated form as

$$\int_0^L b_y(y') \left[\frac{1}{y' - y} + A(y', y) \right] dy' + \int_0^d b_x(y') B(y', y) dy' + f(y) = 0 \quad (0 < y < L) \quad (3a)$$

$$\int_0^L b_y(y') C(y', y) dy' + \int_0^d b_x(y') \left[\frac{1}{y' - y} + D(y', y) \right] dy' + g(y) = 0 \quad (0 < y < d). \quad (3b)$$

The kernels $A(y', y)$, $B(y', y)$, $C(y', y)$, $D(y', y)$, and the functions $f(y)$ and $g(y)$ can be derived from expressions given by Lo (1978).

For a given branch crack length L , the unknowns are the densities $b_x(y)$ and $b_y(y)$ and the extent of opening d . (The density $b_x(y)$ is nonzero only on $0 < y < d$.) Whenever $0 < d < L$, that is, when there is opening over some, but not all, of the interface, then the contact condition $b_x(d) = 0$ must be satisfied (Dundurs and Comninou, 1979). A standard numerical method of solving the singular integral equations via discretization was used (Erdogan, Gupta, and Cook, 1973), with the expected singularity $1/\sqrt{(L-y)}$ built into $b_y(y)$ (and into $b_x(y)$ if $d = L$).

Results and Discussion

In Figs. 2 and 3 the stress intensity factors at the branch-crack tip are plotted as a function of ratio L/a . Stress intensity factors based on a proper accounting of contact at the branch crack faces are plotted as the solid curves. The dashed curves correspond to the forced imposition of the traction-free condition $\sigma_{xx} = \sigma_{xy} = 0$ along the branch crack (as has been generally used in the study of branch cracks), irrespective of possible crack-face overlap. It can be seen that K_I is nonzero for only

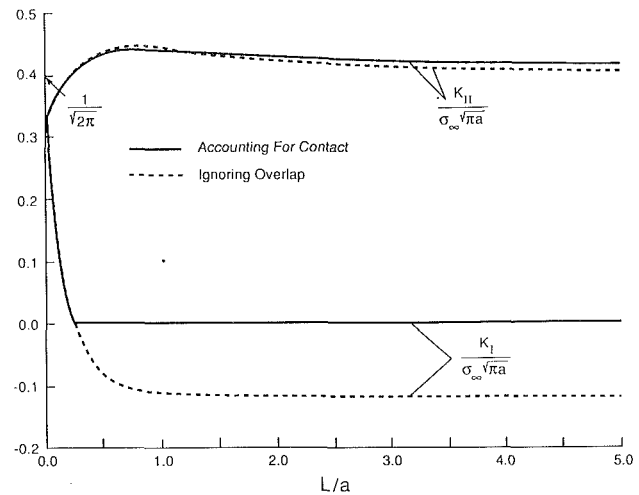


Fig. 2 Branch-crack stress intensity factors for short and moderately long branch cracks

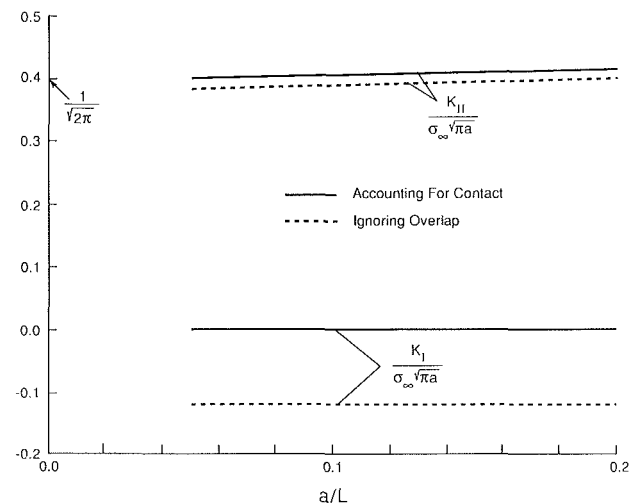


Fig. 3 Branch-crack stress intensity factors for the long branch cracks

a small range of L/a ($0 < L/a < 0.24$), as might be discerned from the work of others in which K_I is found to be negative when L/a is in excess of, roughly, 0.24. There is a somewhat larger range of L/a ($0 < L/a < 0.71$) in which there is some opening of the interface. This means that for $0 < L/a < 0.24$, the branch crack is fully open ($d = L$); for $0.24 < L/a < 0.71$, the branch crack is partially open ($0 < d < L$); and, for $L/a > 0.71$, the branch crack is completely closed ($d = 0$).

It is potentially of great significance that the mode I stress intensity factor becomes zero as L/a increases, whereas an analysis based on a debond which is vanishingly small compared with the main crack would lead one to think that the stress intensity factors are comparable. This significance derives from experimental observations (e.g., Cao and Evans, 1989) that much less energy is required to drive a crack in mode I than in mode II. Hence, much less debonding might be predicted for a given remote loading once it is appreciated that the mode I stress intensity factor is zero.

In general, we find that our results for both stress intensity factors (corresponding to forced imposition of $\sigma_{xx} = \sigma_{xy} = 0$ on the interface) are within 2.5 percent to 6 percent of the tabulated results of Vitek (1977). Kitagawa and Yuuki (1977) presented results only in graphical form. It appears that our results for K_I are within several percent of their results. In the case of K_{II} , however, not only do their results differ by as much as 10 percent from ours and Vitek's, but their K_{II} reaches

a maximum value for L/a being approximately 0.2; by contrast, our K_{II} is maximum for $L/a=0.71$, and Vitek's K_{II} is maximum for L/a between 0.5 and 1.0.)

Neglecting branch-crack contact has an influence on the computed value of K_{II} , though, at approximately 5 percent for large L/a , this is not a substantial effect. With proper accounting for branch-crack contact, one can use an energy release argument to calculate exactly the asymptotic value of K_{II} for the limit as $L/a \rightarrow \infty$. The energy released with a unit extension of the branch cracks is precisely equal to the strain energy of a block of material (of width $2a$) under uniaxial, plane-strain tension σ_∞ . This energy goes to two branch crack tips and is entirely taken up by K_{II} , since $K_I=0$; the result is

$$\lim_{L/a \rightarrow \infty} K_{II} = \sigma_\infty \sqrt{\frac{a}{2}} = \sqrt{\frac{1}{2\pi}} \sigma_\infty \sqrt{\pi a} = 0.3989 \sigma_\infty \sqrt{\pi a}.$$

Our computed values for $K_{II}/\sigma_\infty \sqrt{\pi a}$ at $L/a=10$ and $L/a=20$ are 0.4062 and 0.4013, respectively. Extrapolating linearly to $a/L=0$, we find $K_{II}/\sigma_\infty \sqrt{\pi a}$ approaches 0.3963, a difference of 0.65 percent.

Acknowledgments

This work has been supported in part by the National Science Foundation under grant MSM-8451080, by the General Electric Engine Business Group, and by the Department of Mechanical Engineering, Carnegie Mellon University.

References

- Cao, H. C., and Evans, A. G., 1989, "An Experimental Study of the Fracture Resistance of Bimaterial Interfaces," *Mech. Materials*, Vol. 7, p. 295-304.
- Dundurs, J., and Comninou, M., 1979, "Some Consequences of the Inequality Conditions in Contact and Crack Problems," *J. Elasticity*, Vol. 9, pp. 71-82.
- Erdogan, F., Gupta, G. D., and Cook, T. S., 1973, "Numerical Solution of Singular Integral Equations," *Mechanics of Fracture 1. Methods of Analysis and Solutions of Crack Problems*, G. C. Sih ed., Noordhoff, The Netherlands.
- Kitagawa, H., and Yuuki, R., 1977, "Analysis of Branched Cracks Under Biaxial Stresses," *Fracture 1977*, Vol. 3, D. M. R. Taplin, ed., University of Waterloo, Waterloo, Canada, pp. 201-211.
- Lo, K. K., 1978, "Analysis of Branched Cracks," *ASME JOURNAL OF APPLIED MECHANICS*, Vol. 45, pp. 797-802.
- Vitek, V., 1977, "Plane Strain Stress-Intensity Factors for Branched Cracks," *Int. J. Fract.*, Vol. 13, No. 4, pp. 481-501.

Singularities in p -Type Finite Element Computations: Numerical Experiments on the Locality of Pollution Effects

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Using a recently developed code for three-dimensional p -type finite element computations in elasticity, we have studied, with a series of numerical experiments, the behavior of the solution in the presence of singularities. Worst-case theoretical estimates predict a global pollution of the solution, unless optimally graded meshes are used. On the other hand, we observe that except in an immediate neighborhood of a singularity, and even using uniform meshes, the quality of the solution is not

degraded to any practically relevant extent. This has important practical consequences because it allows modelers to introduce artificial singularities for the purpose of simplifying models. The size of the area around a singularity where the solution is appreciably degraded can be estimated in terms of the minimum effective wavelength of the basis functions used; the latter, in turn, can be related to the element size and the polynomial order used.

Introduction

In p -type finite element methods (e.g., Babuska, Szabo, and Katz, 1981), convergence is reached by systematically increasing the polynomial order of the basis functions used, rather than by mesh refinements as in standard finite element approaches (h -type). An abundant literature has concentrated on error estimates for the h and p -type finite element methods: Babuska (1986), Dorr (1984), Gui and Babuska (1986), Babuska and Suri (1987), Szabo (1986), Tong and Pian (1973), and Zienkiewicz and Craig (1986) are only a small subset of relevant references. The general conclusions are: that the p -method achieves better convergence rates (energy norm versus number of degrees-of-freedom) than the h -method for smooth solutions; that, even in the presence of nonsmooth solutions involving singularities, the convergence rate of the p -method is never worse than the convergence rate of the h -method; and, that optimal convergence rates are achieved, in the presence of singularities, by a combination of the h and p -methods, in which optimally graded meshes around singularities are used.

Although it is often expected (e.g., Schatz and Wahlbin, 1979) that the local accuracy becomes progressively better as the distance from the singularities increases, this is not guaranteed in general. This situation is often referred to as a "pollution effect" (Babuska and Rosenzweig, 1972; Blum and Rannacher, 1988). Thus, conventional wisdom requires that graded meshes be used in the presence of singularities.

In this work, rather than focusing on how to find optimal meshing strategies, we emphasize some of the practical aspects of using a p -type finite element code for elasticity problems. In particular, one often wants to trade optimal meshes for simplified geometries, and it then becomes important to assess the practical consequences of this trade-off from the point of view of achievable accuracy.

In cases where a good description of the singularity is needed (e.g., to estimate stress intensity factors in crack problems, Johnson, 1981), an appropriate grading of the mesh is definitely needed. On the other hand, in models of elastic problems "artificial" singularities (typically due to sharp reentrant corners) are often introduced in order to simplify the model from the geometrical point of view. For example, a mechanical piece might have a reentrant "corner" with a small but finite radius of curvature, and the modeler might not necessarily be interested in computing the stress distribution correctly in the neighborhood of the "corner" because it is known in advance that there is no stress problem in that area. However, this can be done safely only if no practically significant error is introduced by this procedure, except in the neighborhood of the singularity. Functional analytical error estimates cannot be used directly to estimate actual error distributions in realistic problems, and numerical computations are needed.

Using a recently developed p -type finite element code (Carnevali, Morris, and Tsuji, 1990), we have performed a sequence of numerical calculations which address precisely this question. The conclusions of this study are reassuring and indicate that significant errors are only present in the immediate neighborhood of the singularity.

Outside this area no noticeable pollution occurs. The size of this area—like in the familiar situation of a Gibbs phenomenon (e.g., Stakgold, 1979)—is comparable to the minimum wavelength of the basis functions used and can be readily

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Manuscript received by the ASME Applied Mechanics Division, Nov. 17, 1989; final revision, Apr. 15, 1990.